YILI YANG, PhD

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AI Scientist | Geospatial & Remote Sensing

SKILLS

DOMAINS

Remote Sensing
Geospatial Information Science
Machine Learning & Deep Learning
Multimodal Learning
Image Segmentation|Object Detection
Time-series modelling
Earth Foundation Models

TECH STACK

Python|PyTorch
Google Cloud Platform
Google Earth Engine

ENGINEERING

Docker|GitHub Actions
Git|GitHub

EDUCATION

UNIVERSITY OF EDINBURGH

PhD Petrophysics,2021
MSc Geology, 2016
BSc Geoscience, 2015

STATS

PUBLICATIONS

Journal Papers: 13
Conferences: 8
Data Sets: 2

ACADEMIC SERVICES

Invited Talks & Panels: 17
Peer-reviews: 8
Mentees: 9

LINKS

Personal Website
LinkedIn || ORCID
Google Scholar || GitHub

EXPERIENCE

AI SCIENTIST -- GEOSPATIAL & REMOTE SENSING | Woodwell Climate Research Center

Jan 2022 - Present | Falmouth, MA, USA

- **Leadership**: Project lead, pan-Arctic thaw slump mapping with deep learning
- **Create Knowledge**: First circumpolar thaw slump map (19,068 polygons, open at the NSF Arctic Data Center) and the ARTS benchmark dataset
- **Proposals**: Led 8 proposals, contributed to 2; PI on Funding for Climate Solutions 2025-26
- **Share Knowledge**: Machine learning workshop series 2022-2023; 9 mentees, work presented at AGU
- **Partnerships**: Facilitated the Microsoft Planetary Computer partnership (early stage); modelling collaborations with UCONN and ASU
- **Technical Skills**: Zero-shot segmentation on SAM; Arctic-scale inference optimised on GCP

DATA SCIENCE FELLOW | Faculty.AI

Sep 2021 - Dec 2021 | London, UK

- Commercial data-science fellowship, with an industry client placement

DOCTORAL RESEARCHER |ICCR

International Centre for Carbonate Reservoirs
Sep 2016 - Apr 2021 | Edinburgh, UK

- PhD in multiphase flow in porous media; deep-learning segmentation for digital rocks

SELECTED PUBLICATIONS

- Mapping retrogressive thaw slumps using deep neural networks — *Remote Sensing of Environment*, 2023.
- A collaborative and scalable geospatial data set for arctic retrogressive thaw slumps with data standards — *Scientific Data*, 2025.
- Keeping pace with a changing planet: An interactive segmentation framework for refining delineations of dynamic Earth features with the Segment Anything Model — *International Journal of Applied Earth Observation and Geoinformation*, 2026.

AWARDS & GRANTS

FUNDING FOR CLIMATE SOLUTIONS

Funding for Climate Solutions, Woodwell Climate Research Center
$99,749| 2025-2026 | A Generic Climate AI Framework for Multi-domain Time Series Prediction

MULTIDISCIPLINARY COLLABORATIONS

MICROSOFT × WOODWELL

Planetary Computer, permafrost data & compute (early stage)

GOOGLE × WOODWELL

Understanding Permafrosts, AlphaEarth and Satellite Embedding

ESRI × WOODWELL

Understanding Permafrosts and Indigenous Communities